Chapter 6, Problem 64P

Problem

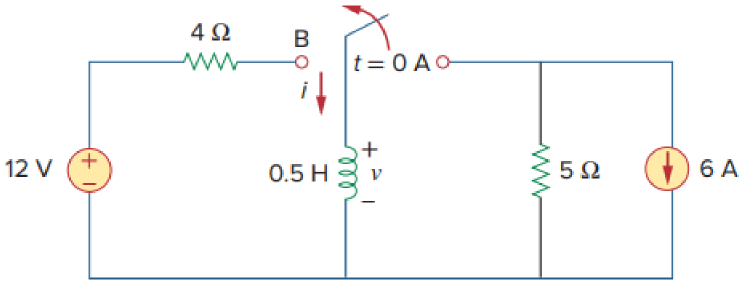
The switch in Fig. 6.86 has been in position A for a long time. At $t = 0$, the switch moves from position *A* to *B*. The switch is a make-before-break type so that there is no interruption in the inductor current. Find:

(a) i(t) for $t > 0$,

(b) *v just after* the switch has been moved to position *B*,

(c) $v(t)$ long after the switch is in position *B*.

Fig. 6.86:



Step-by-step solution

Step 1 of 10

(a)

Refer to the Figure 6.86 in the text book.

At $t < 0\,\text{s}$, the switch is closed at position A.

Draw the following circuit diagram for $t < 0\,\text{s}$.

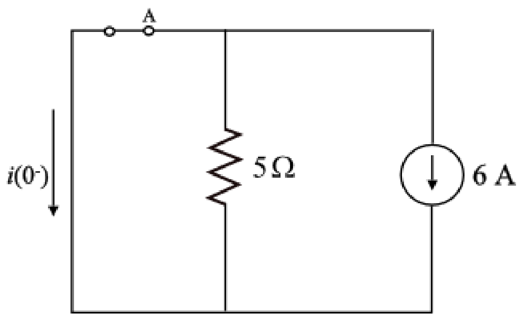


Figure 1

Step 2 of 10

Determine the value of current $i(0^-)$ using Figure 1.

$i(0^-) = -6\,\text{A}$ [Since short circuit]

Step 3 of 10

At $t = 0\,\text{s}$, the switch is closed at position B.

Draw the following circuit diagram for $t = \infty\,\text{s}$.

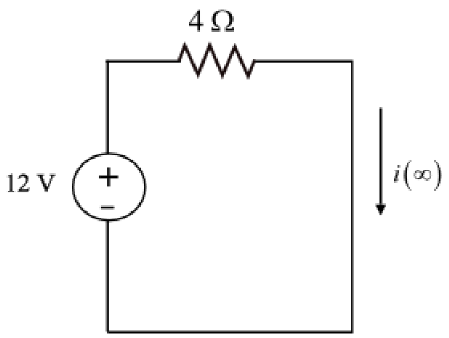


Figure 2

Step 4 of 10

Determine the value of current $i(\infty)$ using Figure 2.

$$i(\infty) = \frac{12\text{ V}}{4\,\Omega}$$
$$= 3\text{ A}$$

Step 5 of 10

Draw the following circuit diagram for $t = 0\,\text{s}$.

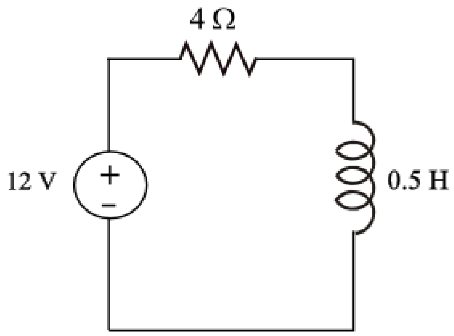


Figure 3

Step 6 of 10

Determine the value of time constnat using Figure 3.

$$\tau = \frac{L}{R}$$
$$= \frac{0.5}{4}$$
$$= \frac{1}{8}\text{s}$$

Step 7 of 10

Determine the current flowing thorough the inductor for $t>0$.

$$i(t)=i(\infty)+[i(0)-i(\infty)]e^{\frac{-t}{\tau}}$$
$$=3+[-6-3]e^{\frac{-t}{\left(\frac{1}{8}\right)}}$$
$$i(t)=3-9e^{-8t}\text{ A}$$

Therefore, the current flowing thorough the inductor for $t>0$ is $\boxed{i(t)=3-9e^{-8t}\text{ A}}$.

Step 8 of 10

(b)

It is known that, the current inductor does not change instantaneously.

Draw the following circuit diagram for $t=0\text{s}$.

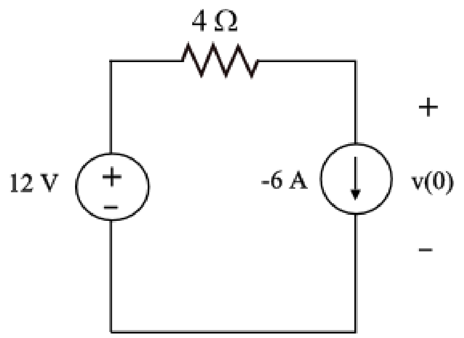


Figure 4

Step 9 of 10

Apply kirchoffs voltage law in Figure 4.

$$-12+(4\Omega)(-6\text{ A})+v(0)=0$$
$$-12-24+v(0)=0$$
$$v(0)=36\text{ V}$$

Therefore, the voltage acrodd indcutor after switch has moved to position B is $\boxed{36\text{ V}}$.

Step 10 of 10

(c)

Determine the voltage across indcutor after the switch is in positin B.

$$v(t)=L\frac{di(t)}{dt}$$
$$=(0.5)\frac{d}{dt}\left[3-9e^{-8t}\text{ A}\right]$$
$$=(0.5)\left[0+72e^{-8t}\text{ V}\right]$$
$$=36e^{-8t}\text{ V}$$

Therefore, the voltage across indcutor after the switch is in positin B is $\boxed{v(t)=36e^{-8t}\text{ V}}$.